

In last month's column, we had briefly mentioned memory networks for question-answering. Memory networks were originally proposed by Jason Weston *et al* in the team's 2014 paper (<https://arxiv.org/abs/1410.3916>). This work was later extended by Ankit Kumar *et al* in the paper, 'Ask Me Anything: Dynamic Memory Networks for Natural Language Processing'. If you are planning to implement a solution for a machine reading comprehension task, both these papers will prove useful.

One of the choices we need to make when we encode the passage of text is the representation granularity. Should we represent the entire passage as a single sequence of words? Or should we first segment the passage into sentences and represent each sentence separately? There are advantages and disadvantages of either choice. Encoding each fact separately allows us to identify the key supporting fact, which potentially contains the answer by matching it with the question. In this case, we can create a fixed length vector representation for each fact and combine these vectors into a matrix. The disadvantage is the loss of information when we compress the fact into a fixed-length vector. On the other hand, representing the entire passage of text as a single sequence allows us to retain word level representation, but is not well suited for key supporting fact extraction. Both the original memory network paper by Jason Weston and the subsequent dynamic memory network paper use the fact level representation. However, their performance has been shown to be poorer compared to passage level sequence representation.

Let us assume that we decide to go with passage level sequence representation. Recall that we decided to form a matrix representation, wherein the LSTM outputs at each time-step (after processing each word in the sequence) were concatenated together. In this case, all words in the passage are given equal weightage in the concatenated matrix representation. However, not all words in the passage are equally important, when our task is to find the answer to a given question.

Given that the answer to a specific question is present only in a specific part of the passage of text, it would be useful to weigh the different parts of the passage with respect to their relevance to the question. This allows us to create a non-uniform weighted representation of the LSTM outputs at each time-step to construct the final matrix representation. In short, we 'attend' to specific parts of the passage of text based on the question, while constructing the encoded passage representation. This idea of selective weighing of different parts of the text is known as the attention mechanism.

To give you a simple example, let us assume that you are shown a picture of people playing football and asked to identify whether there is a red coloured ball in the picture. You would focus your attention not on the human players, but on any images of balls contained in the picture and check what colour the ball is. This idea of choosing some selective sub-set of the entire information is the basis of attention mechanism.

In the context of the machine reading comprehension task, let us assume that you are given a long passage which is about American presidents. You are then asked the question, "What is the birth place of President Obama?" You tend to scan the sentences/passage selectively where the entity 'Obama' appears to find the answer, rather than re-reading the whole text again. This is the phenomenon of attention, whereby we tend to be selective in consuming a large set of information given, to satisfy our specific need.

Attention mechanism has been widely used in neural networks, both in computer vision and NLP. A concise overview of the attention mechanism in neural networks can be found in Chris Olah's blog post at <https://distill.pub/2016/augmented-mnns/>. While we discussed the idea of creating an attentional representation of the passage of text based on the question, it is possible to have bi-directional attention, wherein the question text is also encoded into an attentional representation based on the passage of text. The concept of attention for text was first introduced in the paper 'Neural Machine Translation by Jointly Learning to Align and Translate' by Bahdanau *et al*. The team used the attention mechanism to improve the neural machine translation by selectively focusing on specific words in the source sentence's text, when generating specific parts of the target sentence. Using an attention model enables you to overcome the disadvantages associated with a fixed-length vector representation of the source sentence, which is problematic when translating long sentences. The team showed that its attentional model was able to perform better than phrase-based translation models (based on the hard alignment of source and target phrases).

Attention can either be hard or soft attention. In the context of the machine reading comprehension task, if we decide to select only one sentence out of the passage of text as the key supporting fact based on the question, then it would be a form of hard attention. On the other hand, if we give different weights to different parts of the passage of text without explicitly dropping any part of the passage, then it is a form of soft attention. We will discuss more on attention mechanism in next month's column.

If you have any favourite programming questions/software topics that you would like to discuss on this forum, please send them to me, along with your solutions and feedback, at sandyasm_AT_yahoo_DOT_com. Wishing all our readers happy coding until next month! 

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